
CSE 599 Final Report

Lirui Wang

Department of Computer Science
University of Washington

Abstract

Reinforcement Learning (RL) holds the promise of self-supervised learning from trials and errors, and has shown impressive results in tasks ranging from attaining super-human performance in video-games [33; 6] to learning behaviors for rich tasks and environments [23; 29]. However, compared with supervised learning, RL has from additional challenges such as credit assignments, stability, and explorations, which makes large-scale real-world deployment difficult. Most notably is the poor data inefficiency especially when the policy learning is end-to-end, which motivates the usage of representation learning that has shown large success in supervised learning setting. Moreover, representation learning techniques can also work with the large-scale unsupervised off-policy data. The hope is that the robustness, generalization, and efficiency problems of nowadays RL can be improved with representation learning. In this work, we study several representation and reinforcement learning works that focus on the representation and exploration problem in MDP with low sample complexity and high-dimensional input. We also present a few works that have shown empirical success in robotic applications by connecting representation learning to RL.

1 Introduction

Compared to the standard supervised learning setting, the sequential decision making aspect of Reinforcement Learning (RL) requires the algorithm to tackle problems such as credit assignment, explorations, long-horizon reasoning etc from significantly less supervision signals. While RL can learn from unsupervised self-experiments, but it often struggles with two problems: (1) large sample complexity (2) exploration in huge state space. Such challenge exists in model-based learning, policy-based learning, and value-based learning in RL, and even extend to imitation learning which is more closely related to standard supervised learning. A common strategy to tackle this problem is to leverage the function approximation scheme in the unsupervised collected data to provide a compact representation such that the policy learning can be more efficient, and generalize across states, domains and even tasks in the low-dimensional feature space.

While we observe many success for RL in simulations, in practice RL with deep function approximators require millions of samples to learn a single task. Given that supervised learning setting has empirical risk minimization as an efficient learning method, a natural question for imitation learning is that if demonstrated trajectories for multiple tasks can benefit the sample complexity for future skill Learning. [3] proposes a bi-level optimization scheme where the outer optimization tries to learn the joint representation and the inner optimization learn task-specific skills. The formulation has statistical gains of representation learning for imitation learning, even in the observation-alone setting. A analogous problem for RL is that what is the necessary condition for sample efficiency with regards to representation learning. This problem have been explored in terms of approximation errors but a negative answer is given in [11] even in the known transition model setting. With given good representation, the learning still require exponential lower bounds in terms of planning horizon h for value-based, model-based, and policy-based algorithms given good representations, which

have some further implications in the gap among RL algorithms and some limitations naive usage of representation learning in RL.

The usage of function approximation alone does not resolve the exploration problem in RL when facing complex state and action space. Often practitioners adopt unsupervised learning that collects batch of observation and learn a mapping offline to guide explorations. [3] shows that the exploration driven by unsupervised learning is provably efficient given that the rich observations come from a small number of latent states that the offline learning stage tries to model. On the other hand, [2] suggests that the naive model-based planning approach, where a transition model is learned with maximum likelihood and then used for policy learning, is minimax optimal. The benefit is that it allows for any planning algorithm to be used with the offline model to achieve the low error rate.

From an application perspective, in recent years, we have observed an increasing number of robotic applications that successfully adopt representation learning in RL. Applying representation learning in RL can be roughly classified into four methods: auxiliary loss, state representations, explorations, and unsupervised skill discovery. These applications motivate some theoretical analysis, and the theoretical guarantees also identify the correct setting for the practitioners to adopt the algorithms. In general there are numerous seminal works in both representation and reinforcement literature and there are many open problems yet to be solved, so this survey of some current progress hopes to provide some insights on researchers' attempts to address the challenges in current RL framework, from both a theoretical and application perspective. From a control perspective, other interesting topics involve combining online planning with offline learning to achieve robust performance [31], the generalization effects of overparametrization in learning effective state representation for feedback control [45], and convergence of policy gradient in classical setting such as LQR [16].

2 Preliminaries

2.1 Markov Decision Process

Let $\mathcal{M} = (\mathcal{S}, \mathcal{A}, H, P, R)$ be an Markov Decision Process (MDP) where $\mathcal{S}$ is the state space, $\mathcal{A}$ is the action space whose size is bounded by a constant, $H \in \mathbf{Z}_+$ is the planning horizon, and $P : \mathcal{S} \times \mathcal{A} \mapsto \Delta(\mathcal{S})$ is the transition function that maps state action pair to the next state and $R : \mathcal{S} \times \mathcal{A} \mapsto \mathbb{R}$ is the reward distribution. A fixed initial state s_0 is assumed and a policy $\pi : \mathcal{S} \mapsto \Delta(\mathcal{A})$ maps a state to a distribution over actions. For each $h \in [H] = 1, 2, \dots, H$, we use $\mathcal{S}_h \subseteq \mathcal{S}$ to denote the set of states at level h . The goal is to find a policy that maximize the expected total reward or value function $V^\pi = \mathbb{E}[\sum_{h=0}^{H-1} r_h | \pi]$ over the induced sample trajectories $s_0, a_0, r_0, \dots, s_{H-1}, a_{H-1}, r_{H-1}$. The Q function can be defined as $Q(s, a) = r(s, a) + \mathbb{E}_{s' \sim P(s'|s, a)}[V^\pi(s')]$. The gap between a state-action pair is defined as $\text{gap}(s, a) = \arg \max_{a' \in \mathcal{A}} Q^*(s, a') - Q^*(s, a)$. Similarly, let π^* denote the optimal policy and a policy is said to be ϵ -optimal if $V^\pi \geq V^{\pi^*} - \epsilon$.

2.2 Representation Learning and Block MDP

In practice, we often want the policy π to map from high dimensional state to action, for which representation learning is naturally used. In the block MDP setting [10], $\mathcal{S}$ is a finite unobservable latent state space (referred as underlying MDP), and $\mathcal{X}$ is a possibly infinite observable context space which can be partitioned into $|\mathcal{S}|$ disjoint blocks $\{X_s\}_{s \in \mathcal{S}}$. In this case, we also need the context-emission distribution $q(x|s)$ and decoding functions f_h that maps every observation at level h to its true latent state. The policy in BMDP has the form $\phi = \pi \circ \hat{f}_{[H+1]} : [H] \times \mathcal{S} \mapsto \mathcal{P}(\mathcal{A})$ such that $\phi(h, x_h) = \pi(h, \hat{f}_h(x_h))$. In the stationary MDP case [3] specifically, we often consider the decoding function f as the representation function with bounded norms $\|f(x)\| \leq R$. Softmax linear function policy or argmax are often used in the form $x \mapsto \text{softmax}(Wf(x))$. Denote $\prod^f$ to be the class of policies that use f as the representation function.

2.3 Multi-Task Imitation Learning

We can also add a subscript μ to the MDP to denote multi-task setting. For every task μ , we overload the notation and denote $\pi_\mu^* = (\pi_{1,\mu}^*, \dots, \pi_{H,\mu}^*)$ denote the demonstration trajectories from the expert.

The goal is to first learn representation functions based on which only a new policy can be learned with a few demonstrations on the new task.

2.4 Generative Model

In many settings, our problem is non-episodic, which means that we might have access to a generative model or a sampler, which can provide us with samples $s' \sim P(\cdot|s, a)$ given any pair of (s, a) . Suppose we call our sampler N times at each state action pair and define the empirical model $\hat{P}(\cdot|s, a) = \frac{\text{count}(s', s, a)}{N}$ where **count** counts the number of times the state action pairs (s, a) transitions to s' .

3 Representation Learning for Policy Learning

In this section, we investigate works that investigate the sufficient sample complexity required for RL with near-perfect representations, and how to utilization a learned representation in a multi-task imitation learning setting. The first work proves that for certain family of MDPs, exponential number of samples in horizon are required simply due to the complexity of sparse reward. The second work shows that for both standard or observation-alone setting, sufficiently large demonstration tasks and expert trajectories can guarantee that a new task from the same distribution can be efficiently learned.

3.1 Is a Good Representation Sufficient for Sample Efficient Reinforcement Learning? [11]

While previous works have focused on the sufficient conditions to achieve polynomial sample complexities, the requirements on lower bounds of representations that suffice for sample efficient RL are less explored. Assume that the transition model P is known, which implies that without exploration difficulty, sparse reward itself is hard enough. The goal of this work is to show that with approximation error $\delta = \Omega(\sqrt{\frac{H}{d}})$ where d is the dimension of the representation (which can be huge) and H is the horizon, value-based and model-based methods would still require exponential number of samples (in terms of H) to learn near-optimal policy. For the policy-based method, it requires exponential number of trajectories to learn a near-optimal policy, even with a strictly positive margin for a perfect linear predictor on a representation. The proof is based on construction of a binary-tree hard instance with known P and R . There are H levels and 2^h states in level h . There's only one state with reward $R = 1$ and all others zero. Intuitively, to find a near-optimal policy, the agent needs to enumerate all states in the tree, which is exponential in horizon H . We restate the theorem below and brief the proof idea, and refer interested readers to the original paper for the detailed proof and technical details.

Theorem 1. (Exponential Lower Bound for Value-based Learning (Theorem 4.1 [11])). There exists a family of MDPs with $|\mathcal{A}| = 2$ and a feature extractor ϕ that satisfy policy completeness (assumption 4.2 [11]) such that any algorithm that returns a $1/2$ -optimal policy with probability 0.9 needs to sample $\Omega(\min\{|\mathcal{S}|, 2^H, \exp(d\delta^2/16)\})$ trajectories.

The theorem of exponential lower bound for linear transition model has the same statement but with the requirement of approximate linear MDP. Note that the bounds can be generalized to $|\mathcal{A}| > 2$ and also linear feature extractor is a subclass of more complex function classes such as neural networks. Moreover, the assumption on value-based lower bound is bounded approximation error in the L-infinity sense which is stringent yet necessary, but the same uniform bound on the linear transition matrix can be a weak condition that might be enforced to achieve polynomial sample complexity.

The specific proof is based on the the upper bound $\epsilon - \text{rank}(I_n) \leq O(\log n/\epsilon^2)$, which can be rewritten as $\|I_{2^H} - \Phi\Phi^T\|_\infty \leq \epsilon$. Note that Φ can be approximately unitary and the row of Φ can be taken as random unit vector and still satisfy the bound in the assumption. Moreover, any learning that can suboptimal needs to discover the sparse reward in level $H - 1$, thus requiring exponential number of samples. Therefore, it implies that almost all feature extractors induce a hard MDP instance.

Since previous work have shown a polynomial upper bound for exactly perfect representation, the near-perfect representation pays the gap of exponential complexity. Similarly, in supervised setting where $H = 1$ or the imitation learning setting, learning a linear classifier requires only polynomial number of samples to achieve near-optimal policy. Therefore, the exponential sample complexity dependence on H shows the difficulty of RL.

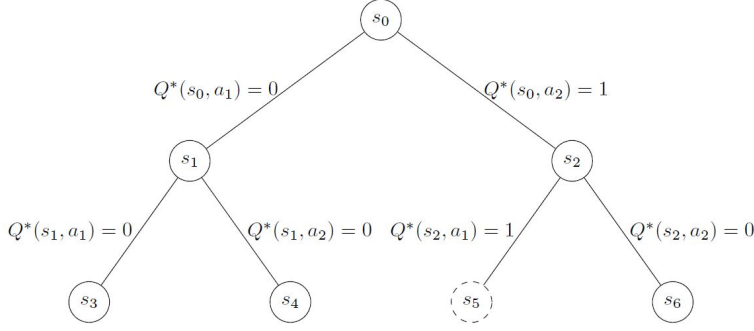


Figure 1: [11]. The binary tree hand instance with $H = 3$. In this case only $\bar{r}(s_5) = 1$ (dashed) and $\bar{r}(s) = 1$ all other states. This instance can be shown to satisfy the assumptions on near-perfect representations and still requires exponential trajectory samples to find s_5 given a known transition model.

Theorem 2. (Exponential Lower Bound for Policy-based Learning (Theorem 4.3 [11])). There exists an absolute constant Δ_0 , such that for any $\Delta < \Delta_0$, there exists a family of MDPs with $|\mathcal{A}| = 2$ and a feature extractor ϕ that satisfy optimality gap with $\rho = \frac{1}{2 \min\{H, d\}}$ and the optimal linear policy has a margin Δ (Assumption 4.5 [11]), such that any algorithm that returns a $1/4$ -optimal policy with probability at least 0.9 needs to sample $\Omega(\min\{2^H, 2^d\})$ trajectories.

This theorem is even more pessimistic compared with the previous one in the sense that even with gap and margin in addition to a perfect representation, the agent still requires exponential number of samples. Compared this with near-perfect representation assumption in value-based method, it implies that learning the optimal Q function can be much harder than learning the optimal policy. The proof involve some geometric techniques and requires $d = \Theta(H)$ for the usage of curse of dimensionality. The high level idea is to create a set $\mathcal{N}$ such that $|\mathcal{N}| = (1/\Delta)^{\Omega(d)}$ and show that there exists a hyperplane L that separates p and the rest and all vectors in $\mathcal{N}$ has distance at least Δ to L . Then the vectors in $\mathcal{N}$ can be used as features and the exponential sample bound from the binary tree instance still holds even if there are margins.

3.2 Provable Representation Learning for Imitation Learning via Bi-level Optimization [3]

Representation learning can be very useful to extract common structure from multiple tasks and improve new skill learning and reduce sample requirements. This paper analyzes the statistical gain of multi-task imitation learning setting when there are multiple tasks and many expert demonstrations for each task, which resembles the multi-task representation learning for supervised learning setting. The main idea of the paper is the following bi-level optimization $\min_{\phi \in \Phi} L(\phi) = \mathbb{E}_{\eta \in \mu} \min_{\pi \in \Pi} \ell^\mu(\pi)$

where ℓ^μ is the inner loss function uses to penalize π from π^* . The logistic function is used in the paper as a surrogate to the 0 – 1 loss similar to [37]. Φ is the class of representation functions, η denote the task distribution, μ denotes sample distribution for a task μ . In practice, only samples $x^{(1)}, \dots, x^{(T)}$ from μ are used, so the empirical version of L is $\hat{L}(\phi) = \frac{1}{T} \sum_{i=1}^T \ell^{x^{(i)}}(\pi^{\phi, x^{(i)}})$ where

$\pi^{\phi, x} = \arg \min_{\pi \in \Pi^\phi} \ell^x(\pi)$ corresponds to a task specific policy that uses a fixed representation ϕ . Two assumptions are made to prove the theorem: (1) The expert policy is deterministic for every task μ . (2). There is a representation $\phi^* \in \Phi$ such that for every μ and $\pi_\mu \in \Pi^{\phi^*}$ $\pi_\mu(s) - \pi_\mu^*(s) \geq 1 - \gamma$ for some $\gamma < 1/2$. This assumes the existence of a expressive representation that can approximate expert policies. Under these assumptions, a PAC-style guarantee can be given as (See Corollary 5.1 in [3]).

Theorem 3. If the number of tasks satisfy $T \geq c_1 \max\{\frac{H^4 R^2 \log(|\Phi|)}{\epsilon^2}, \frac{H^4 \ln(4/\delta)}{\epsilon^2}\}$, and the number of expert trajectories per task satisfies $n \geq c_2 \frac{H^4 R^2 |\mathcal{A}|}{\epsilon^2}$ for some small constant c_1, c_2 , then with

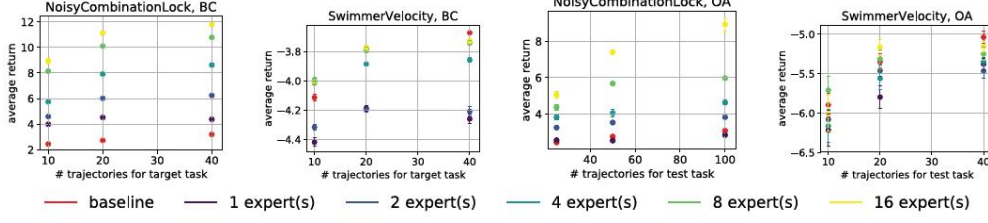


Figure 2: [3]. Different experiments for standard BC and observation-alone BC setting. 1 - 16 experts are used to compare imitation learning based on learned representation to the baseline method (without representation learning). The results align with the theory that more tasks/experts and more trajectory demonstrations help the performance.

probability $1 - \delta$,

$$\mathbb{E}_{\mu \sim \eta} \mathbb{E}_{x \sim \mu^n} J_{\mu}(\pi^{\hat{\phi}, x}) - \mathbb{E}_{\mu \sim \eta} J_{\mu}(\pi_{\mu}^*) \leq H^2 \gamma + \epsilon$$

The above bound says that as long as there are enough tasks to learn a representation from Φ and sufficient samples per tasks to learn a linear policy, the policy will have small average cost on a new task from η . The sample requirement dependence changes from $\log(|\Phi|)$ without access to tasks and representations to $|\mathcal{A}|$, which might be more sample efficient when the representation function class is large.

The proof has two main steps. The first step bounds the errors due to use of samples. The policy $\pi^{\hat{\phi}, x}$ that is learned on samples $x \sim \mu$ is evaluated on the distribution μ . The second step connects the empirical loss from ℓ^{μ} to J_{μ} . This has been established in previous work [37] to show that a policy that has small regression loss would also do well in task and causes an error term $H^2 \epsilon$.

Note that the idea can be extended to observation alone setting when the experts' actions are not observed but only the states. In this case, a per-level policy is used and a discriminator class [40] is assumed to contain the true value function. Under realizability assumptions on value and policy functions, the inner optimization problem in the observation-alone setting can be modelled as a min-max problems. In the observation-alone imitation learning case, the conclusion still holds (Fig. 1) that with sufficient number of tasks and demonstrations, the learning of new tasks require less sample complexity.

4 Model-Based and Unsupervised Learning Oracles for Efficient RL

In this section, we investigate two works that utilize optimization oracles and unlabelled environment samples for efficient RL methods. The first work focuses on using unsupervised learning algorithm to learn representations that can achieve good exploration efficiency to learn latent space policy. The second work shows that any plug-in model-based planning approaches on an empirical MDP is minimax optimal to learn a good policy in the true MDP with sample size sublinear in the model size.

4.1 Provably Efficient Exploration for RL Using Unsupervised Learning [17]

There is a substantial body of work on provably efficient tabular RL and function approximations are often adopted for huge observation space. Often assumptions are made on transition models or the underlying latent state MDP to show polynomial sample requirements. This work lifts some of the assumptions and is motivated by applications that combine unsupervised learning for RL explorations [41].

The goal of this work [17] is to find a near-optimal policy with sample complexity polynomial to the cardinality of the latent state space. Instead, the access to an unsupervised learning algorithm $\mathcal{ULO}$ and a no-regret tabular RL algorithm $\mathcal{A}$ are required, and both routines should return suboptimal results with polynomial samples. The algorithm defines ϵ -suboptimality, $1 - \delta$ probability to output the latent decoding and find the optimal policy, and sample size n that in the limit leads to no representation error with f_h up to some permutation. More details can be found in definition 1 and 2 [17]. The high-level algorithm (shown in Algorithm. 1) combines the unsupervised learning oracles

and RL algorithms as black boxes to collect observations and simulate enough episodes with the latest decoding functions.

Algorithm 1: A Unified Framework for Unsupervised RL

Input: BMDP $\mathcal{M}$, unsupervised learning oracle $\mathcal{UL}\mathcal{O}$, RL Algorithm $\mathcal{A}$, constants N, L, ϵ, δ

for $i = 0, \dots, N$ **do**

 Clear the memory of $\mathcal{A}$ and restartl

for $k = 0, \dots, K$ **do**

 obtain π^k from $\mathcal{A}$

 Obtain a trajectory $\tau^k, f_{[H+1]}^k$ from $\text{TSR}(\mathcal{UL}\mathcal{O}, \pi^k, B)$

 Update the algorithm $\mathcal{A}$ with τ^k

end

 Form policy $\phi^m := \pi^{K+1} \circ f_{[H+1]}^{K+1}$ from π^{K+1} and encoding function $f_{[H+1]}^{K+1}$

end

Run each ϕ^n for L episodes and get the average rewards per episode V_{ϕ^n}

Output the best-performance policy $\phi \in \arg \max_{\phi \in \phi^{[N]}} V_{\phi}$

TSR(Trajectory Sampling Routine) is a subroutine that takes policy and the latest-learned decoding functions f_h to collect new trajectories, and fix labels together with previous trajectories in the cache. Each label update step requires some uniform trajectories sampling and labels from previous decoding function. After learning, the new decoding function is used to obtain a trajectory as the final sample. This procedure is repeated and under certain assumptions on the sample complexity of the oracles $\mathcal{UL}\mathcal{O}$ and the blackbox RL algorithm $\mathcal{A}$. The following theorem holds for Algorithm. 1.

Theorem 4. Suppose $\mathcal{UL}\mathcal{O}$ and $\mathcal{A}$ are (ϵ, δ) correct with $\text{poly}(|\mathcal{S}|, |\mathcal{A}|, H, \frac{1}{\epsilon}, \log(\delta^{-1}))$ sample complexity. Then Algorithm. 1 outputs a policy ϕ such that with probability at least $1 - \delta$, ϕ is an ϵ -optimal policy for the BMDP, using at most $\text{poly}(|\mathcal{S}|, |\mathcal{A}|, H, \frac{1}{\epsilon}, \log(\delta^{-1}))$ trajectories.

4.2 Model-Based RL with a Generative Model is Minimax Optimal [2]

The natural question to ask for model-based planning is if the naive “plug-in” approach non-asymptotically, minimax optimal given a fixed sample size. This work considers the sample and computation complexity of obtaining an ϵ -optimal policy given access to a generative model. It shows that any efficient planning algorithm in the plug-in model constructed with N samples can provide an ϵ -optimal policy in the true MDP.

The simplest model-based RL method first builds the maximum likelihood estimate of the transition model in the MDP M from observations and then find an optimal policy in this empirical MDP $\hat{M}$. Previous work have shown that using $\tilde{O}(\frac{|\mathcal{S}||\mathcal{A}|}{(1-\gamma)^3\epsilon^2})$ samples from the generative model is sufficient for an ϵ -optimal value function. The same sample complexity for policy holds for the model-based algorithm in this work and is known to be not improvable. Slightly different from previous work, this work requires the optimization oracle takes input an MDP M and returns a suboptimal policy satisfying: $\|V_M^\pi - V_M^*\|_\infty \leq \epsilon_{\text{opt}}$ and $\|Q_M^\pi - Q_M^*\|_\infty \leq \epsilon_{\text{opt}}$.

The key challenge of understanding the sub-optimality of the returned policy $\hat{\pi}$ (learned in $\hat{M}$) in the true MDP M is that the value estimate is not an unbiased estimator. A usual approach of uniform convergence would result in a crude bound scaling as $|\mathcal{S}|^2$. In this work, a black-box oracle is utilized to show the following theorem.

Theorem 5. Suppose $\delta > 0$ and $\epsilon \in (0, (1-\gamma)^{-1/2})$. Let $\hat{\pi}$ be any ϵ -optimal policy for $\hat{M}$, i.e. $\|Q_M^\pi - Q_M^*\|_\infty \leq \epsilon_{\text{opt}}$. If $N \geq \frac{c\gamma \log(c|\mathcal{S}||\mathcal{A}|(1-\gamma)^{-1}\delta^{-1})}{(1-\gamma)^3\epsilon^2}$, we have

$$Q^{\hat{\pi}} \geq Q^* - \epsilon - \frac{9\epsilon_{\text{opt}}}{1-\gamma} \quad \text{and} \quad V^{\hat{\pi}} \geq V^* - \epsilon - \frac{9\epsilon_{\text{opt}}}{1-\gamma}$$

with probability at least $1 - \delta$, where c is an absolute constant and $\gamma \geq 1/2$.

Thus, the theorem shows that if ϵ_{opt} is made suitably small, then we will find an $O(\epsilon)$ sub-optimal policy with N samples for each (s, a) pair. The total sample complexity would be $|\mathcal{S}||\mathcal{A}|N$ which is known to be unimprovable in the regime of $\epsilon \in [0, 1]$ from previous work.

Moreover, there are some specific implications on the computation complexity of sparse model-based planning. The value iteration approaches requires $O((1 - \gamma)^{-1} \log \epsilon_{opt}^{-1})$ iterations with each iteration takes $O(|\mathcal{S}||\mathcal{A}|N)$ time. Policy iteration on the other hand can take similar iteration complexity but requires solving a linear system of size $|\mathcal{S}|^2$ each iteration. Overall the analysis is agnostic to the specific model-based approaches used in the oracle. The relatively simpler proof of the theorem (off by a factor of $\sqrt{(1 - \gamma)^{-1}}$ proceeds as follows. The component-wise bounds is first established $Q^\pi \geq Q^* - \|Q^\pi - \hat{Q}^\pi\|_\infty - \|\hat{Q}^\pi - \hat{Q}^*\|_\infty - \|Q^{\pi^*} - Q^*\|_\infty$ where the middle term is the optimization error. For the other generalization terms, the proof follows [4] by some Bellman properties and union bound with Hoeffding bounds over the state space.

5 Applications

In practice, deep RL methods is not entirely "deep" in the sense that the policy or value functions are often parametrized with two or three-layer networks with relatively small width. When the problem is from high-dimensional input to action, or "pixel to torque", an over-parametrized network is then used to map from observation to some low-dimensional features. These features are often called latent variables of representations that attempts to summarize the perception input of the system. The major challenge for real-world deployment of current RL is an unstable training procedure with inefficient learning and explorations. Moreover, real-world tasks typically have sparse reward structure and robustness requirements. Still, while theory has achieved many progress, we have also observed empirical success in applying RL in many tasks. This section would survey robotic applications of representation learning in RL can be roughly classified into four methods [14]. In addition, we also present a few works on related few-shot learning and model-based learning in RL.

5.1 Auxiliary Loss

Standard supervised learning typically offers strong generalization guarantees for in-distribution data. An easy-to-implement way to regularize representation learning in RL is to supervise prediction tasks for information of interests, which ideally requires no external labelling. In sparse reward setting, the agent has little supervision signals yet need to address problems such as credit assignment, which makes the learning landscape difficult even for low-dimensional tasks. [25] shows that using unsupervised auxiliary tasks can improve efficiency and task success. Specifically, they augment the sparse reward RL problem with auxiliary rewards such as pixel and state change as well as a reward prediction task. They obtain improvements in both performance and efficiency in simulation domain. The multi-objective learning is to maximize total rewards: $\arg \max_{\theta} \mathbb{E}_{\pi} [R] + \lambda_c \mathbb{E}_{\pi_c} [R^{[c]}]$, where c denotes auxiliary reward task, R is the cumulative discounted reward and λ is balancing parameter. Moreover, state reconstruction tasks [18], state estimation loss [46], and contrastive loss [38] have also been used to improve representation learning. These methods leverage the power of neural net in optimizing over multiple objectives. Its unsupervised and simplicity nature also aligns well with the theory work and has shown effectiveness in many RL applications.

5.2 State Representation

The power of deep net to compress useful semantic information in lower-dimension embedding is very attractive in deep RL setting. [44] proposes to use linear Gaussian controller in the embedding space [21] learns environment dynamics directly from pixels and use it for planning, and [20] uses variational Bayes methods together with sequential models to encode low dimensional latent from high dimensional input and complex dynamics. On the other hand, [1; 30] study the theory of approximate state abstractions for MDP to reduce task complexity. Specifically, the paper shows the existence of different types of approximate state aggregation functions that achieve polynomially-suboptimal bounds. These works have shown strong ability to capture the complex, high-dimensional input in relatively simple domains.

5.3 Exploration

How to conduct efficient and useful explorations is a longstanding topic in AI. [41] uses count-based method with hash tables and unsupervised learning for explorations. [8] study intrinsically motivated RL for explorations in a large scale by curiosity that is measured by prediction error of the

environment dynamics. [24; 39] models exploration as a maximization of information gain about the agent’s belief of environment dynamics, from an information-theoretic perspective. Closely related to the next section, latent variables are also used as representations of exploration strategy in hierarchical RL frameworks [22]. It would be interesting to see the role of latent variables to improve the efficiency and safety limitations in real world setting.

5.4 Unsupervised Skill Discovery

Good state representations provide an interesting direction to discover novel skills. [22; 15; 22; 32] use self-supervised methods to learn in a multi-task settings with latent variables representing each task and skills to accomplish the tasks. Typically an agent first learns skill embedding from large-scale unlabelled data and further inform downstream policy learning with the latent space prior. Generative modelling such as variational inference or normalizing flows, and probabilistic formulation of RL [27] are often used in the framework. This results in better policy generalization, transferable skill representations, and systematic explorations compared to action space explorations. In general, these representation methods often work with large-scale off-policy exploration data, so they have less data requirements compared to onpolicy methods.

5.5 Model-Based Policy Learning

Represent the transition model of the complex world with a neural network often provides orders of magnitude sample efficiency improvement. Guided Policy Search (GPS) [28; 34] splits the problem into a “control (C) step” that finds multiple simple local policies for different initial states and a “supervised (S) step” that optimizes the global policy. [13] uses a deep net to generate visual foresight for manipulation tasks. Most deep model-based RL works use the learned model as a black-box simulator to generate samples [26]. Iterative Learning Control [7] uses the actual rollouts to remove the bias of model rollouts. [31] combine MPC planning online with model and terminal value learning offline. [43] benchmarks model-based controllers in simulation domains. More recently [36; 9] attempt to combine policy learning with model learning. While it’s not trivial to acquire an exact model of the real world dynamics, these methods are orders of magnitude more efficient in sample data.

5.6 Few-Shot and Meta Learning

Few-shot and meta-learning (or learning to learn) approaches have a substantial body of literature work. They more directly consider the problems of using representations learned from many other tasks in order to reduce the sample complexity for learning new task. [5; 12] show that it is possible to gain sample benefits in both meta and few-shot learning. The new skill generalization problems are often modelled as an online learning problem where the samples come in sequence and is typically not assumed to be i.i.d. A more thorough survey on first-order meta learning can be found [35; 42]. The application sides of few shot learning have been focusing on image classification and simulation tasks [19]. It is very appealing to have a framework that do not have to re-learn every single skill with millions of new agent experience.

6 Conclusion

In conclusion, we present a survey on the connection between representation learning and reinforcement learning from both theoretical and application perspectives. These methods are proposed to analyze and tackle the challenge of sample complexity and exploration problems in RL when facing high dimension state spaces, which is the central motivation for end-to-end RL applications. Specifically [11; 3] studies the sufficient of good representations to improve sample complexity, and a framework to learn representation for imitation learning setting with bi-level optimization. [3; 2] study methods to leverage unsupervised data to explore or to plan efficiently. We also present a short survey on applications of representation learning in RL that have shown very promising directions.

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